A Formal Theory of Definition

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Abstract

This paper presents a formal theory of definition based on the fundamental distinction between a conceptual space (C), containing thinkable concepts, and an experiential space (E), containing empirical entities. We formalize the notion of a definiendum (definiendum $\in C$) and its dependency set (dependencies $\subseteq C \cup E$). Definitions are classified into three types: analytical (precise, linguistic, establishing $C-C$ relations), exemplary (linguistic examples, inductive, establishing $C-C$ relations), and ostensive (empirical pointing, inductive, establishing $C-E$ relations). We introduce the concept of a definition system ($\mathcal{S}$) as a set of definitions whose dependency graph is a directed acyclic graph. The initial nodes of this graph are primitives (P), which can be either conceptual (P_C) or experiential (P_E). A system is grounded if it contains experiential primitives ($P_E \neq \varnothing$), which necessitates the use of ostensive definitions. Within this framework, an assertion is a statement about conceptual relations in a specific system. This theory can be applied to explain communication failures (mismatched systems), logical fallacies (shifting systems), and the grounding problem in artificial intelligence, among other issues.

Keywords: Definition, Conceptual space, Experiential space, Analytical definition, Exemplary definition, Ostensive definition, Symbol grounding problem

1 Theory

1.1 Conceptual Space and Experiential Space

The conceptual space (C) and the experiential space (E) are two disjoint sets:

$$C \cap E = \varnothing.$$

Conceptual space (C): Elements in this space (concepts) can be thought, such as "table", "sun", "triangle", "line", "existence", or "truth".

Experiential space (E): Elements in this space (physical entities, events, etc.) exist in the empirical world. We cannot directly present elements of the experiential space in language; we can only *refer* to them through concepts. Examples include the table in front of you, the sun above, an action you just performed, the current scene, or a present feeling.

Note the distinction: "sun" itself is a concept, while the entity it refers to is an element of the experiential space.

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1.2 Definiendum and Dependency Set

Let d represent a definition.[1] Let definiendum_d denote its definiendum (the term being defined), and dependencies_d denote the *set* of elements (from C or E) upon which it depends. For example, for the definition

$$d = \text{"A bachelor is a person who has completed undergraduate studies and received the corresponding degree"},$$

$\text{definiendum}_d = \text{"bachelor"}$, and $\text{dependencies}_d = \{\text{"person"}, \text{"complete"}, \text{"studies"}, \ldots\}$.

The definiendum must be a concept (elements in the experiential space do not require and cannot be defined):

$$\text{definiendum}_d \in C.$$

The dependency set is a subset of the union of the conceptual and experiential spaces:

$$\text{dependencies}_d \subseteq C \cup E.$$

The definiendum cannot be an element of its own dependency set:

$$\text{definiendum}_d \notin \text{dependencies}_d.$$

1.3 Classification of Definitions

Definitions can be classified in various ways, each offering a different perspective and application. This paper classifies definitions into analytical definitions, exemplary definitions, and ostensive definitions.

Analytical definition: Defines a term through a *precise* and *equivalent* description using pure language. The audience[2] can (in principle) accurately determine and judge the definiendum based on an analytical definition. Analytical definitions include genus-differentia definitions, operational definitions, recursive definitions,[3] and enumerative definitions. Analytical definitions establish relationships between elements within the conceptual space. For example, "A triangle is a closed planar geometric figure formed by three line segments connected end-to-end" establishes a relationship between the concepts "triangle", "three", "line segment", "connected", etc.

Exemplary definition and **ostensive definition**: These define a term by providing *incomplete samples*, forcing the audience to perform *incomplete induction* (akin to machine learning). This induction process is open-ended (different audiences will induce at least slightly different boundaries), contrasting with the closed nature of analytical definitions.

Exemplary definition: Provides examples using pure language, establishing relationships between elements in the conceptual space. For instance, listing several awkward

[1] Note the difference between *defining* a concept and *describing* one: a definition provides the boundaries of the concept, whereas a description states its characteristics.

[2] The "audience" herein refers to any individual (human, machine, etc.) receiving the definition through any sensory modality.

[3] In a recursive definition, although the definiendum is linguistically self-referential, it does not belong to the dependency set. For example, in Peano's definition of natural numbers, the dependency set includes concepts such as "0" and "successor" but not "natural number" itself. The audience only needs to know the concepts in the dependency set beforehand to understand the definition.

scenarios and stating, “This is called ‘embarrassment’ ”; or defining “existence” by showing several of its uses.

Ostensive definition: Defines a concept by pointing to elements in the experiential space, thereby establishing a mapping between conceptual and experiential elements. It relies on non-linguistic means of indication. For example, pointing to a horse and saying, “This is a ‘horse’ ”[4]; pointing to a color swatch and saying, “This is ‘red’ ”; saying in a specific awkward situation, “This is called ‘embarrassment’ ”; or after witnessing sarcasm, “That person’s behavior just now was ‘sarcasm’ ”.

Analytical and exemplary definitions do not involve the empirical world, whereas ostensive definitions establish the relationship between concepts and the empirical world.

$$\forall \text{ analytical or exemplary definition } d, \text{dependencies}_d \subseteq C;$$

$$\forall \text{ ostensive definition } d, \text{dependencies}_d \cap E \neq \varnothing.$$

1.4 Definition Set and Definition Dependency Graph

A **definition set** $\mathcal{D}$ is a set of definitions.

We can construct a **definition dependency graph** $G(\mathcal{D}) = (V_\mathcal{D}, \mathcal{E}_\mathcal{D})$ by taking all elements involved in $\mathcal{D}$ as nodes. Directed edges are drawn from elements in the dependency set to the definiendum.

$$\text{Node set: } V_\mathcal{D} = \{\text{definiendum}_d \mid d \in \mathcal{D}\} \;\cup\; \bigcup_{d\in\mathcal{D}} \text{dependencies}_d;$$

$$\text{Edge set: } \mathcal{E}_\mathcal{D} = \{(x, c) \mid \exists d \in \mathcal{D} \text{ s.t. } c = \text{definiendum}_d \text{ and } x \in \text{dependencies}_d\}.$$

In the dependency graph, a sequence of nodes formed by traversing against the direction of the edges (i.e., along the direction of dependency) is called a **dependency chain**:

$$(v_0, v_1, v_2, \dots) : \;\forall i \geq 0,\; (v_{i+1}, v_i) \in \mathcal{E}_\mathcal{D}.$$

1.5 Definition System and Primitives

If, in a definition set $\mathcal{D}$, all definienda are distinct, the dependency graph $G(\mathcal{D})$ is a directed acyclic graph (DAG), and no infinite dependency chains exist, then the set $\mathcal{D}$ and all elements $V_\mathcal{D}$ involved constitute a **definition system** $\mathcal{S} = (V_\mathcal{D}, \mathcal{D})$.

Definition systems are not unique. For example, system A might define “horse” using an ostensive definition, while system B uses a genus-differentia definition. These are different systems.

In the dependency graph of a definition system, nodes with an in-degree of 0 are called **primitives**:

$$P(\mathcal{S}) = \{v \in V_\mathcal{D} \mid \text{in-degree}(v) = 0\}.$$

A non-trivial definition system must have primitives:

$$\mathcal{S} \neq (\varnothing, \varnothing) \Rightarrow P(\mathcal{S}) \neq \varnothing.$$

Primitives can be classified into:

[4] Clearly, defining by a *single* indication is highly ineffective, as different audiences will induce vastly different results. Ostensive definitions typically involve *multiple* indications. Furthermore, this implies “horse” *can* be defined ostensively, not that it *must* be.

- **Conceptual primitives**: $P_C(\mathcal{S}) = P(\mathcal{S}) \cap C$ (undefined initial concepts)
- **Experiential primitives**: $P_E(\mathcal{S}) = P(\mathcal{S}) \cap E$ (elements from the experiential space)

Clearly, $P_C(\mathcal{S}) \cup P_E(\mathcal{S}) = P(\mathcal{S})$.

If experiential primitives exist ($P_E(\mathcal{S}) \neq \varnothing$), the system is called a **grounded definition system**. A grounded system must contain ostensive definitions. A system without ostensive definitions will have only conceptual primitives.

If all primitives are experiential ($P(\mathcal{S}) = P_E(\mathcal{S})$), the system is a **fully grounded definition system**. This is equivalent to $P_C(\mathcal{S}) = \varnothing$ (all concepts involved in the system are defined).

1.6 Assertion

Within a given definition system $\mathcal{S} = (V_{\mathcal{D}}, \mathcal{D})$, one can make an **assertion** (A), which essentially states a relationship between elements in the conceptual space:

$$A = \Psi(c_1, c_2, \ldots, c_k),$$

where $c_i \in C \cap V_{\mathcal{D}}$, and Ψ is a k-ary predicate.

Assertions must be made based on an (explicit or default) definition system. An assertion in a vacuum is meaningless.

If an assertion is intended to describe relationships in the empirical world, it must be based on a grounded definition system. For example, the assertion "The Earth revolves around the Sun" states a relationship between the *concepts* "Earth" and "Sun". The assertion, combined with the definitions (in this case, ostensive definitions for "Earth" and "Sun"), jointly describes a relationship between entities in the empirical world (the physical Earth and the physical Sun).

Definition systems with conceptual primitives often use basic assertions (i.e., axioms) to state the relationships between them. Axioms are undecidable. For example, in Euclidean geometry, the axiom "Through any two points, there is exactly one line" states a relationship between the conceptual primitives "point" and "line".

2 Examples of the Theory's Application[5]

2.1 Discussions Should Be Based on a Shared Definition System

In daily life, people often communicate without explicitly stating their definitions. This usually works because the definition system is *default* and (nearly) identical among participants (any differences are irrelevant to the current discussion).

However, sometimes communication breaks down ("people talking past each other"). This may be because differences in definition systems affect the topic at hand, requiring participants to *clarify* their definitions. A common case is when people from different cultural backgrounds have vastly different definitions for abstract concepts (e.g., "politeness").

A scenario where different systems are intentionally used is a debate competition. Modern debates are often for spectacle rather than truth-seeking. The affirmative and

[5]This section is not the primary focus of this paper.

negative sides adopt different definitions for the core concepts in the topic. For example, in a debate on "In life, which is more important: career or love?", the two sides will define "more important" differently, enabling a vigorous debate.

Many questions are matters of definition, not discussable issues. For example, "What is justice?" is a definitional question; the real, discussable question is "What should a social system be like?". Another example: "Is God a being that looks human?" This depends on the definition of God. The discussable question is, "If the definition of 'God' includes the condition 'looks human', does God exist?"

2.2 Discussions Should Be Based on a Fixed Definition System

Discussions should be based on a fixed definition system; otherwise, it leads to the logical fallacy of "moving the goalposts".[6]

2.3 The Grounding Problem in Large Language Models

Traditional large language models (LLMs) are trained solely on text. They learn the relationships between concepts (e.g., "red") and other concepts (e.g., "apple", "blood", "color"). However, all definitions in their definition system are *analytical* or *exemplary*. The definition system $\mathcal{S}_{\text{text}}$ of such a model is *ungrounded*.

In contrast, multimodal large models (or, further, embodied intelligence) use *ostensive definitions*. The model connects the concept "red" in conceptual space with elements in the experiential space (e.g., a picture of a red apple, sensor-captured images). Only such a "grounded" artificial intelligence can truly "understand" the world.

References

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[6]Note the difference between "moving the goalposts" and "equivocation". "Equivocation" occurs within a *single* definition system $\mathcal{S}$ by exploiting the ambiguity of a word (e.g., the same word "light" corresponding to at least two different concepts, c_1 = bright and c_2 = not heavy). "Moving the goalposts", in contrast, involves abandoning the original definition system $\mathcal{S}_{\text{A}}$ mid-argument and replacing it with a new system $\mathcal{S}_{\text{B}}$ that is more favorable to the argument (e.g., in $\mathcal{S}_{\text{A}}$, "bright" is defined as "not dim, allowing one to see objects", whereas in $\mathcal{S}_{\text{B}}$, it is defined as "sufficiently illuminated to read books").